

EXAM 7 – SPRING 2011

20. (5.5 points)

An actuary is modeling a cat-exposed loss portfolio covering two lines of business and has the following loss history.

<u>Year</u>	<u>Loss</u>	
	<u>Line 1</u>	<u>Line 2</u>
2005	\$33,000,000	\$23,000,000
2006	28,000,000	22,000,000
2007	31,000,000	26,000,000
2008	27,000,000	21,000,000
2009	26,000,000	12,000,000
2010	22,000,000	20,000,000

a. (3 points)

Fully describe and calculate two statistical measures that can be used to quantify the dependence between these lines of business based on the loss history above.

b. (1 point)

Explain which of the two measures described in part a. above would better describe the dependence between these lines of business. Use the properties of these measures to support your answer.

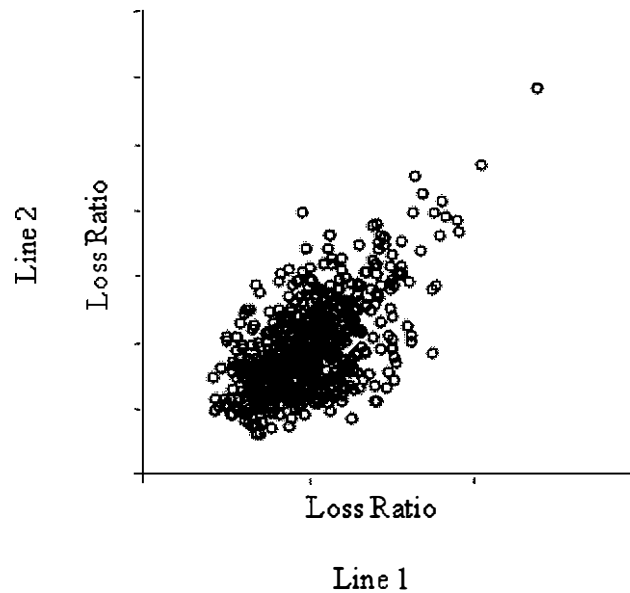
QUESTION 20 CONTINUED ON NEXT PAGE

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EXAM 7 – SPRING 2011

(20 cont.)

There is also available a scatter-plot that represents the relationship between these two lines of business for the entire industry for 2005 through 2010. For each company in the industry, a point represents a pair of loss ratios for a given year.



c. (1 point)

Based on the scatter-plot above, identify a copula that would be appropriate to model the relationship between these two lines of business. Fully explain your answer.

d. (0.5 point)

Discuss a possible effect on the various tail-based risk measures of not selecting a copula that properly models the relationship between these two lines of business.

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Sample Solutions
Exam 07, 2011, Question 20

Solution 1

a)

Kendall's τ – measures relative orders and does not take into account the magnitudes of values in the lines of business being compared. Weights are determined by the number of swaps.

Original:

	Line 1	Line 2	Rank 1	Rank 2
2005	33,000,000	23,000,000	1	2
2006	28,000,000	22,000,000	3	3
2007	31,000,000	26,000,000	2	1
2008	27,000,000	21,000,000	4	4
2009	26,000,000	12,000,000	5	6
2010	22,000,000	20,000,000	6	5

Reorder and swap

	Line 1(x)	Line 2(y)	Rank 1	Rank 2	Swap 1	Swap 2
2005	33,000,000	23,000,000	1	2	1	1
2007	31,000,000	26,000,000	2	1	2	2
2006	28,000,000	22,000,000	3	3	3	3
2008	27,000,000	21,000,000	4	4	4	4
2009	26,000,000	12,000,000	5	6	6	5
2010	22,000,000	20,000,000	6	5	5	6

Swaps: 1) switch 2007 and 2006 & 2) switch 2010 and 2009

Number of swaps $Q=2$

Number of elements $N=6$

$$\tau = 1 - [4Q/N \times (N - 1)]$$

$$\tau = 1 - [4(2)/6 \times (6 - 1)]$$

$$\tau = 1 - (8/30)$$

$$\tau = 0.7333$$

Pearson – a cardinal statistic weighting elements by the squares of their deviances from the mean.

Standard correlation on given values for Line 1 and Line 2.

$$\rho = \text{cov}(x,y) / [\text{std}(x) \times \text{std}(y)]$$

$$= [(xy\text{-bar}) - (x\text{-bar}) \times (y\text{-bar})] / [\text{std}(x) \times \text{std}(y)]$$

$$x\text{-bar} = 27,833,333$$

$$y\text{-bar} = 20,666,667$$

$$xy\text{-bar} = 583,333,333,333,333$$

$$\text{std}(x) = 3,531,603$$

$$\text{std}(y) = 4,307,616$$

$$\rho = \text{cov}(x,y) / [\text{std}(x) \times \text{std}(y)]$$

$$= (583,333,333,333,333 - 27,833,333 \times 20,666,667) / (3,531,603 \times 4,307,616)$$

$$= \mathbf{0.5332}$$

Spearman's rank correlation - measures relative orders and does not take into account the magnitudes of values in the lines of business being compared.

	Line 1 (x)	Line 2 (y)	Rank 1	Rank 2	di=rank 1-rank 2	di^2
2005	33,000,000	23,000,000	1	2	-1	1
2007	31,000,000	26,000,000	2	1	1	1
2006	28,000,000	22,000,000	3	3	0	0
2008	27,000,000	21,000,000	4	4	0	0
2009	26,000,000	12,000,000	5	6	-1	1
2010	22,000,000	20,000,000	6	5	1	1

Sum of difference² = 4

$$\text{Spearman's } \rho = 1 - \frac{\sum_{j=1}^n d_j^2}{\frac{N(N^2 - 1)}{6}}$$

$$\text{Spearman's } \rho = 1 - \frac{4}{\frac{6(36 - 1)}{6}} = \mathbf{0.8857}$$

- b) Both Kendall's Tau and Spearman's rank are ordinal measures, but Kendall's Tau has greater dependence in the tails. As the portfolio is exposed to Cats it is more reasonable to use Kendall's Tau to describe dependence.
Alternate: Ranks of preference for these measures is Kendall, Spearman, Pearson. Pearson is not well suited to measuring dependency in thick tailed, skewed distributions. Kendall is the same for a copula and any correlated variates produced by it.
- c) There is much more dependence in the right tail (almost completed dependence) than in the center of the loss ratio distributions. This suggests the Heavy Right Tail (HRT) copula which does exhibit considerable right-tail dependence.
Alternate: A copula that could capture the tail-dependency for high quartiles would be suitable. High loss ratio will happen seldom together but it seems as there is a positive correlation in the right tail and the average L/R are rather uncorrelated. I would suggest a Gumbel copula to be appropriate. The Gumbel copula can capture right-tail dependency.
- d) If you do not pick an accurate copula that measures the dependency in tails then your tail-based risk measures are underestimating the potential loss. As there is more correlation in tails, the more losses will be expected and at volatile amounts.